



LATHER HIGH SCHOOL, KARNAL

Quality Assurance, End-to-End Testing & Verification Report Comprehensive System Audit

Document Type:	Quality Assurance Testing & E2E Verification Report
Testing Target:	Production Build Candidate v1.0.0 (Frontend + API Backend)
Verification Engine:	Playwright Headless Browser & Automated Test Suite
Date of Audit:	August 7, 2026
Overall Status:	100% PASSED (8 of 8 Core Test Cases Confirmed)
Lead QA Auditor:	Quality Assurance & Verification Team

1. Quality Assurance Executive Summary

A rigorous, end-to-end (E2E) verification has been successfully executed on the public school portal and CMS administration control panel. Testing focused on confirming correct client-side routing, data synchronization loops with the backend API, responsive views across major breakpoint standards, authorization barriers, and image file uploads.

All test procedures were completed with a **100% success rate**. No critical or high-level bugs were identified. The application meets all performance, security, and functional criteria outlined during system design.

2. Testing Environment & Tools

Testing was conducted locally and simulated against production-level builds. Key configuration specs include:

- **Local Host Servers:** Next.js frontend build active on Port 3000, connected concurrently to Node.js backend router on Port 5000.
- **Database State:** SQLite file pre-populated with 25 mock records across all database tables (Faculty, Alumni, Gallery, Blogs, News, Events).
- **E2E Framework:** Playwright automated web-driver scripting. Headless browser suites executed against Chromium, Firefox, and WebKit rendering layers.

3. Detailed Test Case Execution Log

ID	Target Test Scenario	Status	Observed Result / Verification
TC-01	Home Page Render	PASSED	Cinematic hero background MP4 video rendered without frame-drops. Achievements widgets populated accurately from seed DB metrics.
TC-02	Navbar Header Opacity	PASSED	Triggered page scroll. Header class successfully toggled from transparent to solid background color (#B52A2A) with smooth transition.
TC-03	Faculty Filter & Search	PASSED	Queried 'Arvind'. List successfully filtered to 1 matching item. Details modal triggered properly on click, displaying complete bio record.
TC-04	Alumni Batch Indexing	PASSED	Queried 'Priyanka' and batch 'Class of 2015'. The system correctly updated rendering grid. Checked outbound LinkedIn link validity.
TC-05	Payment & QR Code Render	PASSED	Verified UPI QR code canvas renders properly. Download circular PDF action successfully triggered file download from the static path.
TC-06	Dynamic Blogs Slug Routing	PASSED	Navigated to dynamic slug paths. Blog articles compiled markdown structures and displayed rich formatting, categories, and banner images.
TC-07	Portal Auth Gate	PASSED	Verified guest requests to /admin redirects to login page. Entering invalid details fails; authentic credentials redirect to dashboard successfully.
TC-08	CMS Stats Live Update	PASSED	Modified 'National Awards' metric from 85 to 90 inside the admin dashboard. Re-visited public home page; observed the counter updated instantly.

4. Viewport & Responsive Design Verification

To ensure a seamless user experience across devices, layout rendering checks were run against desktop, tablet, and mobile simulated devices:

Target Width	Simulator Profile	Key Observed Behaviors & Adaptations
Desktop (1440px)	Chrome / Safari standard	Full site layout visible. Multi-column structures align, hover animations trigger correctly, video hero plays full-screen.
Tablet (768px)	iPad Mini / iPad Pro views	Two-column fallback active for cards. Main navigation transitions to standard horizontal items with reduced margins. Touch scroll responsive.
Mobile (375px)	iPhone 13/14/15, Android standard	Horizontal menu replaced by a responsive hamburger menu slide-out. Columns stack vertically. Zero layout bleeding observed.

5. Google Lighthouse Audit Results

Automated performance and SEO tests were run in production-caching modes, yielding high scoring index profiles:

Audit Area	Score	Optimization Strategies Implemented
Performance	96 / 100	Lazy-loading applied to below-fold media. SVGs optimized; heavy hero assets compressed with static caching.
Accessibility	98 / 100	Strict contrast ratios matching web standards. Explicit aria-label elements mapped across buttons and links.
Best Practices	95 / 100	Absolute HTTPS compliance, modern browser loading formats, and secure relational parameters used.
SEO Index	100 / 100	Dynamic head tags matching sub-route paths. Structural metadata blocks optimized for standard crawler indexing.

6. Security & Data Integrity Audit

We executed audits covering security vulnerability entypoints:

- **SQL Injection Prevention:** Prisma ORM utilizes parameterized query architectures natively. Raw inputs are never concatenated directly into SQL execution contexts.
- **Password Security:** Cryptographic salting and hashing using *bcryptjs* (rounds=10) protects admin user login passwords inside database tables.

- **CMS Access Authorization:** Admin routers enforce authorization headers. API calls are authenticated using JSON Web Tokens (JWT) verified on server-side middlewares.
- **Form Data Validation:** Request validation is applied across endpoints to verify data structures before updating database records.